

Generate equivalent ratios

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$. Copy or complete the first step.

2. Simplify $36:48$.

3. Paint is mixed red to white in the ratio $3:5$. How much white is needed with 18 ml red?

4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$. Copy or complete the first step.
Answer 16.
2. Simplify $36:48$.
Answer $3:4$.
3. Paint is mixed red to white in the ratio $3:5$. How much white is needed with 18 ml red?
Answer 30 ml.
4. Miro says: Miro adds the same number to both parts to make an equivalent ratio. Is this correct? Explain.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.

Generate equivalent ratios

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

6. Check this result using an inverse, estimate or known fact: Paint is mixed red to white in the ratio $3:5$. How much white is needed with 18 ml red?

2. Simplify $36:48$.

7. Prove or explain your reasoning: There are 8 red and 12 blue counters. Write red to blue as a ratio.

3. Paint is mixed red to white in the ratio $3:5$. How much white is needed with 18 ml red?

8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.

4. Solve this independently and show every stage: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Simplify $36:48$.

10. Write and solve a similar question to: Paint is mixed red to white in the ratio $3:5$. How much white is needed with 18 ml red?

Teacher answers and checking prompts

1. Complete the equivalent ratio $7:4 = 28:\underline{\hspace{1cm}}$.
Answer 16.
2. Simplify 36:48.
Answer 3:4.
3. Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer 30 ml.
4. Solve this independently and show every stage: Complete the equivalent ratio $7:4 = 28:\underline{\hspace{1cm}}$.
Answer 16.
5. Use a second method or representation for: Simplify 36:48.
Answer Expected result: 3:4. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer Expected result: 30 ml. A valid check must be shown.
7. Prove or explain your reasoning: There are 8 red and 12 blue counters. Write red to blue as a ratio.
Answer 8:12, which simplifies to 2:3. The explanation must show why the method works.
8. Identify and correct this misconception: Miro adds the same number to both parts to make an equivalent ratio.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Generate equivalent ratios

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: There are 8 red and 12 blue counters. Write red to blue as a ratio.

6. Create a new example that tests the same idea as: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?

2. Prove the answer to this question, rather than only calculating it: Complete the equivalent ratio $7:4 = 28: \underline{\hspace{1cm}}$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Simplify 36:48.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 30 ml.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: There are 8 red and 12 blue counters. Write red to blue as a ratio.
Answer 8:12, which simplifies to 2:3. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Complete the equivalent ratio $7:4 = 28:__$.
Answer Expected result: 16. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Simplify 36:48.
Answer Expected result: 3:4. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 30 ml.
Answer One valid reconstruction is: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro adds the same number to both parts to make an equivalent ratio.
Answer Equivalent ratios come from multiplying or dividing both parts by the same factor.
6. Create a new example that tests the same idea as: Paint is mixed red to white in the ratio 3:5. How much white is needed with 18 ml red?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.